



Detection of crowdedness in bus compartments based on ResNet algorithm and video images

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Abstract

The crowding in bus is an important factor affecting passenger satisfaction and bus dispatching level. However, how to use video images to detect crowding accurately is a difficult problem. In this paper, firstly, an image sample library is established based on the evaluation standard of crowding in bus, which contains 16346 sample images. Then, Local Binary Pattern (LBP) and Gray Level Co-occurrence Matrix (GLCM) are used to extract the texture features of the image in bus. Then, a rough classification method of crowding based on Support Vector Machine (SVM) is proposed. At the same time, in order to improve the accuracy of rough classification of crowding, the optimization effects of grid search algorithm, particle swarm optimization algorithm and genetic algorithm on SVM parameters are compared. The results show that the optimization effect of genetic algorithm is the best, and the accuracy rate is 93.20%. Finally, for the problem that the SVM method is not ideal in the fine classification of crowding, this paper proposes a new method based on ResNet. SGD, Adadelta and Adam are selected to optimize the parameters of ResNet model. The accuracy of the optimal Adam algorithm reaches 96.22%, which effectively solves the problem of the fine classification of crowding in bus.

Keywords Bus · Crowdedness · Video images · Support vector machine · ResNet

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1 Introduction

In recent years, in order to alleviate traffic congestion, cities are vigorously developing public transport, and buses have become the main body of urban public transport. However, the current bus operation and service levels still need to be improved. The problem of passenger congestion often occurs in buses, which greatly affects the operation level of public transport and public riding experience, and also affects the attractiveness of buses to the public. Therefore, it is urgent to carry out the research of congestion detection to provide reference data for bus dispatching and route planning, so as to improve the comfort and safety of passengers.

At present, there are still few research on crowding detection in bus at home and abroad, and the related research focuses on the estimation of crowd density. Therefore, this paper designs a method for crowding detection in bus by using the crowd density estimation method with video images as reference, which mainly based on pixel statistics, texture analysis and deep learning.

The method based on pixel statistics was first proposed by Davies et al. [9]. Its principle is to use the linear relationship between the number of foreground pixels or edge pixels and the crowd density to estimate the crowd density. Chow et al. [5] extracted three kinds of pixel statistical features of crowd image, including foreground pixel proportion, background pixel proportion and edge pixel number, and then sent these three kinds of pixel statistical features into hybrid neural network to estimate crowd density. Chow et al. [4] added the human template matching module to the algorithm in the previous paper and replaced the hybrid neural network with RBF neural network. These two improvements greatly improved the accuracy of crowd density estimation and real-time response capabilities of the algorithm. The advantages of the method based on pixel statistics are that the calculation is small and easy to operate, but the effect is often poor when there is crowd occlusion. Therefore, there are few related research on this method.

The method based on texture analysis can solve the problem of crowd occlusion, so the effect of crowd density estimation is better than that based on pixel statistics. It performs well in all kinds of density estimation and becomes a popular method in crowd density estimation. Marana et al. [17] first proposed a method based on texture analysis, extracted texture features by Gray Level Co-occurrence Matrix (GLCM) method, and realized crowd density classification using self-organizing map neural network. After that, Marana et al. [18] respectively used four methods such as GLCM, line segmentation, Fourier analysis and fractal dimension to extract texture features, and then used three classifiers to carry out classification experiments and obtained the accuracy rate under different combinations.

Image texture feature extraction is an important part of the method based on texture analysis. Haralick et al. [10] first proposed the GLCM method in the 1970s and proposed 14 quadratic statistics calculated from the GLCM. The method considers that the spatial distribution of pixels can reflect the texture information of the image, so it can reflect the texture information by counting the joint distribution density of pixels that meet the specified conditions. Ojala et al. [19] proposed a local binary pattern operator with simple calculation and strong robustness to grayscale changes and proved the reliability of the method by using two sets of texture image data sets. Comer et al. [6] proposed to use autoregressive model to describe image texture features, used multi-resolution Gaussian autoregressive (MGAR) model for pyramid representation of the observed images, and assumed multi-scale Markov Random Field model for the class label pyramid. Azencott et al. [1] proposed to use spatial

filter to describe image texture features and defined the distance between textures for texture classification based on window Fourier filter.

After texture feature extraction, the selection of classification algorithm is also very important, which directly affects the effect of crowd density estimation. Cover et al. [8] first proposed k-nearest neighbor classification algorithm. The main idea of the algorithm is to take the most frequent category among k most similar data as the classification result. Simard et al. [22] proposed to use decision tree classifier and multi-scale texture method to extract tropical vegetation cover information from global rainforest map, that is, to distinguish land cover types. Cortes et al. [7] proposed an SVM classifier for optical character recognition. Support Vector Machine classification algorithm has high classification accuracy and is the most widely used. It can well solve the problems of small samples, nonlinearity, high dimensionality, over-learning, and local minima. SVM conceptually realized the nonlinear mapping of input vector into high-dimensional feature space, in which linear decision-making surface was constructed to realize classification. Zhao et al. [32] extracted the color features and texture features of road image, and used SVM classifier to classify, and realized the recognition of road state. In order to improve the recognition accuracy and generality, the kernel function factor and penalty factor of SVM were optimized by using grid search algorithm and particle swarm optimization algorithm, which effectively solved the problem of road state recognition in mixed or different video scenes. Zhao et al. [31] took the improved HU invariant distance feature as the vehicle logo image feature, and used SVM algorithm to recognize the vehicle logo, at the same time, gray wolf algorithm was used to optimize the parameters of SVM algorithm, and good vehicle logo recognition effect was achieved.

With the development of various deep learning models in the field of computer vision, crowd density estimation methods based on deep learning are also emerging. Zhang et al. [28] proposed a deep convolution neural network for group counting and population density estimation. During the training process, the network was alternately trained with the two related learning goals of group count and group density by switching training methods. However, this method needs manual marking to correct the perspective effect, which requires a lot of manpower and has certain subjectivity. Zhang et al. [29] proposed a multi column convolution neural network to realize population counting and density estimation. Each column in the multi column convolution neural network has convolution kernels of different sizes, which can extract the multi-scale features of the crowd, and effectively improve the accuracy rate of population statistics and density estimation. In addition, the size adaptive Gaussian filter was proposed to eliminate the perspective effect. Sam et al. [20] improved the multi column convolution neural network and proposed a multi column convolution neural network with selection function. The network divides the input image into nine non overlapping image blocks, and then selects appropriate convolution neural network for different crowd image blocks, which can not only alleviate the perspective effect, but also improve the efficiency and accuracy of the network. However, the effectiveness of this method largely depends on the performance of the classifier, and it is difficult to get a better classifier. Sindagi et al. [23] proposed a context pyramid convolution neural network composed of a global context estimator, a local context estimator, a density map estimator and a fusion convolution neural network. The network adds context information to the multi column convolution neural network, effectively improving the accuracy rate of population statistics and the quality of population density map. Li Baiping et al. [16] proposed to use the GoogLeNet model for real-time crowd density estimation and to intercept video frames at certain intervals from continuous videos and feed them into the GoogLeNet model to extract deep features and

classification, so as to obtain corresponding crowd density levels. The methods based on deep learning have higher accuracy and computational complexity. The network in the first three papers is mainly designed for crowd density estimation of single image, so the network structure is more complex, and the parameters are huge, which makes it difficult to apply to real-time crowd density estimation. In the fourth paper, a more lightweight network is introduced to reduce the computing cost and realize real-time crowd density estimation. To sum up, it is very important to choose the appropriate network model when using deep learning method to estimate the real-time crowd density, which should consider both the accuracy and the calculation cost.

Reviewing the above literature, we found that the problems and development trend of population density estimation method are as follows:

(1) Using texture analysis and deep learning method to estimate population density is the future development trend. (2) The selection of image features is the key to the method based on texture analysis. (3) SVM and other classification algorithms have been widely used, and the use of appropriate optimization algorithm can help to improve the accuracy.

Therefore, this paper proposes a rough classification method of crowding based on support vector machine and a fine classification method of crowding based on ResNet, which can effectively solve the problem of crowding detection in bus. Firstly, the fusion of LBP and GLCM is used to extract the 12-dimensional texture feature vector of the image in bus, and then the parameter optimized SVM classifier is used to realize the rough classification of crowding. The fine classification method uses ResNet model for feature extraction and classification to realize the fine classification of crowding. In this paper, a large number of bus image samples are used to test the rough classification and fine classification methods, and the ideal crowding detection effect is obtained, which verifies the effectiveness of the proposed method. The technical route of the paper is shown in Fig. 1.

2 Establishment of image sample library in bus

2.1 Evaluation standard of crowding in bus

Congestion is a subjective concept, which is influenced by personal factors such as body shape, personality characteristics and behavior habits, as well as external factors such as space size and temperature, at present, there is no uniform and clear definition. Congestion is a qualitative problem, but it is necessary to formulate its quantitative standards in the establishment of the crowding in bus image sample library and subsequent crowding detection research. The service level of standing passenger area is divided into six levels A~F in the Transit Capacity and Quality of Service Manual (TCQSM2013) [3] by the Transportation Research Board of the United States. The classification is more detailed. This paper takes it as the fine classification standard of crowding, which can provide a more detailed reference for public transport dispatching and public transport planning departments. However, most of the passengers don't know the specific evaluation criteria when providing the information of crowding for them. If there are many grades, it is not conducive for the passengers to grasp the specific congestion situation intuitively. Therefore, it is only necessary to provide an intuitive description of the crowding. On the basis of the above fine classification criteria, this paper proposes a rough classification standard of crowding in bus, which is shown in Table 1.

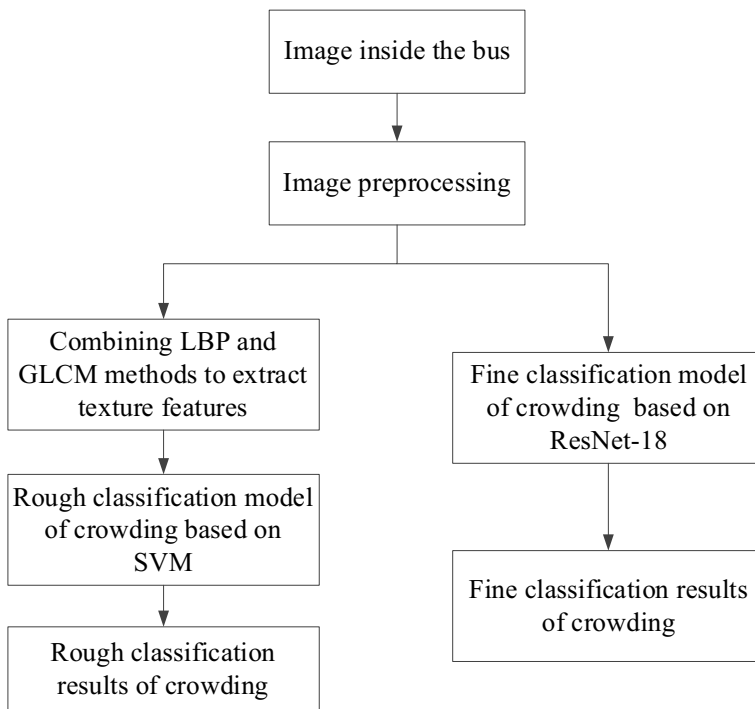


Fig. 1 Technical route of the paper

2.2 Image sources in bus

In this paper, the video images taken by the front camera inside the bus are selected for crowding detection research. Considering the high similarity of adjacent video frames in the bus, video frames are extracted from the video at intervals of 25 frames as sample images. Some sample images are shown in Fig. 2. In order to protect the privacy of passengers, the images shown in this paper have covered the passenger's face and vehicle's information.

According to the fine classification standard of crowding, the collected sample images in bus are classified into six categories, and a sample library of images in the

Table 1 Rough classification standard of crowding in bus

Crowding	Standing passenger area (m ² /person)	Description of passenger crowding
Comfortable	≥ 1.00	Passengers can move freely in the carriage, and most or all passengers have seats
Normal	[0.40, 1.00)	There is more space left in the carriage, and the standing passengers have no physical contact
Crowded	[0.20, 0.40)	Standing passengers have more physical contact and feel a little crowded
Very crowded	< 0.20	Standing passengers have frequent physical contact and feel very crowded, which affects getting on and off



Fig. 2 Image samples collected by camera in bus: **a** comfortable; **b** normal; **c** crowded; **d** very crowded

bus is established, which contains 16346 sample images, including 3956 class A samples, 2917 class B samples, 2405 class C samples, 1841 class D samples, 2225 class E samples and 3002 class F samples, subsequent studies will select the required number of sample images.

3 Texture feature extraction of image in bus

3.1 LBP principle

The principle of LBP Operator is to compare the gray value of the center pixel and other pixels in the neighborhood of a pixel, if the gray value of the pixel is greater than the gray value of the central pixel, it is set to 1, otherwise it is set to 0, get a set of binary numbers describing the local texture features. In this way, the binary sequence of all pixels in the image is calculated, and then convert the binary number into decimal number. Finally, the texture feature vector is obtained by histogram statistics.

In this paper, LBP image is obtained by only using LBP Operator to calculate the image pixels. The pixel values of each point in the LBP image represent the LBP value reflecting the local texture structure of the image, instead of the gray value. Subsequently, the macro distribution of the micro texture structure, that is, the global texture features, is obtained by using the GLCM method on the basis of the LBP image. In this way, local and global texture features can be well combined. An example of LBP image of four kinds of crowding sample images is shown in Fig. 3.

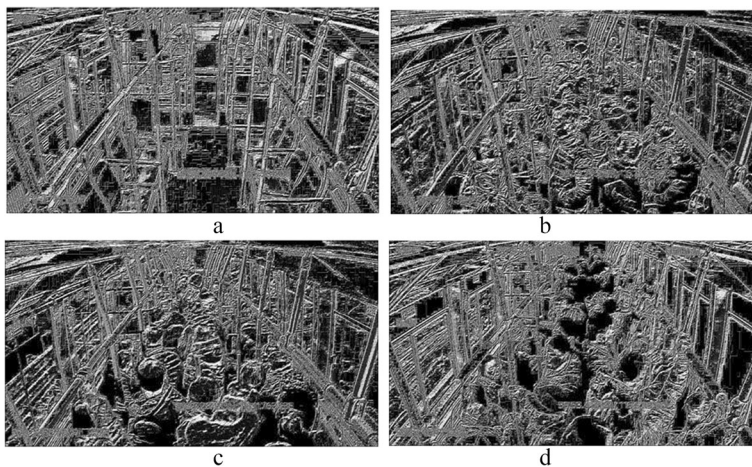


Fig. 3 LBP feature map: **a** comfortable; **b** normal; **c** crowded; **d** very crowded

3.2 GLCM theory

The GLCM method is based on the spatial correlation of the gray levels of image pixels and realizes the description of texture features by studying the spatial correlation of two pixels separated by a certain distance in the image.

Let $f(x, y)$ be a grayscale image of size $M \times N$, (x_1, y_1) and (x_2, y_2) are two pixels in the image with a distance of d in the θ direction, $x_2 = x_1 + d \times \cos\theta$, $y_2 = y_1 + d \times \sin\theta$, the definition of the gray-level co-occurrence matrix is shown in Eq. (1).

$$P(i, j, d, \theta) = \# \left\{ (x_1, y_1), (x_2, y_2) \in M \times N \mid f(x_1, y_1) = i, f(x_2, y_2) = j \right\} \quad (1)$$

In the above formula, $\#\{X\}$ represents the number of elements in set X , d represents the distance between two pixels, θ represents the angle between the line of two pixels and the positive direction of the horizontal axis of the coordinate.

In practical applications, the secondary statistics calculated on the basis of the gray level co-occurrence matrix are usually used as the feature quantity of texture analysis, before calculating the secondary statistics, it is necessary to normalize the gray level co-occurrence matrix according to Eq. (2).

$$p(i, j) = \frac{P(i, j, d, \theta)}{R} \quad (2)$$

In the above formula, R represents the sum of all elements in the gray level co-occurrence matrix.

Haralick [10] proposed 14 kinds of secondary statistics, in this paper, four commonly used secondary statistics that are easy to calculate and have low correlation are selected as the texture characteristics of the image in the bus.

(1) Energy

$$ASM = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} [p(i,j)]^2 \quad (3)$$

Energy is also called Angular Second Moment (ASM), used to reflect the thickness of the texture and the uniformity of the pixel gray value distribution. When the energy value is small, it indicates that the image pixel gray value distribution is more uniform and the texture is finer; when the value is large, it indicates that the image pixel gray value distribution is uneven and the texture is rough.

(2) Contrast

$$Con = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} (i-j)^2 p(i,j) \quad (4)$$

Contrast is a secondary statistic that reflects the depth of image texture grooves and image clarity. When the contrast value is small, it means that the image definition is low and the image texture groove is shallow; when it is large, it means the image is clearer and the image texture groove is deeper.

(3) Entropy

$$Ent = - \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} p(i,j) \log p(i,j) \quad (5)$$

Entropy is a secondary statistic used to reflect the complexity or non-uniformity of image texture. When the entropy value is large, the image texture is messier; when it is small, the image texture is relatively regular and uniform.

(4) Inverse Difference Moment

$$IDM = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} \frac{p(i,j)}{1 + (i-j)^2} \quad (6)$$

Inverse Difference Moment is a secondary statistic that reflects the degree of local change in image texture. When the value of the Inverse Difference Moment is large, it means that the texture of different regions in the image has a small change; when the value is small, it means that the texture of different regions in the image has a large change.

When constructing the gray level co-occurrence matrix, we need to select construction parameters such as direction and distance. This paper randomly selects 30 images from the four types of crowding samples in the image sample library in the bus for experiment, take the number of image frames as the abscissa, compress the image gray level to 16, take $d = 1$, and

compare and analyze the changes of each feature value in the four directions. The experimental results are shown in Fig. 4.

By observing and analyzing the experimental results, it can be found that the change of the feature quantity when $\theta = 90^\circ$ is obviously different from the other three directions, especially the change of contrast and inverse moment. In the other three directions, the change trend of the characteristic value curve when $\theta = 45^\circ$ and $\theta = 135^\circ$ are basically the same, and the numerical value is also relatively close. The characteristic value change when $\theta = 0^\circ$ is somewhat different from the former two and the difference of numerical value is also large. To sum up, this paper chooses the three directions $\theta = 0^\circ, 45^\circ, 90^\circ$ to calculate their respective gray level co-occurrence matrix and its secondary statistics and combine them to form 12-dimensional texture feature vectors.

In this paper, the above samples are tested to select the appropriate distance, take $\theta = 0^\circ$, and compress the number of gray levels to 16. Compare the changes of the four eigenvalues under different values. In order to maintain the texture properties to the maximum extent. In this paper, the values of d are selected as 1, 4, 8, and 16 respectively for experiments [25]. The experimental results are shown in Fig. 5.

From the experimental results in the above figures, we can see that the change trend of the feature quantity of the sample image when the d value is different is basically the same, but the size of the feature quantity is different. In addition, when the value of d is large, the amount of calculation is small, but the loss of texture information is also large; on the contrary, when the value of d is small, the amount of calculation is large, but the texture information remains more

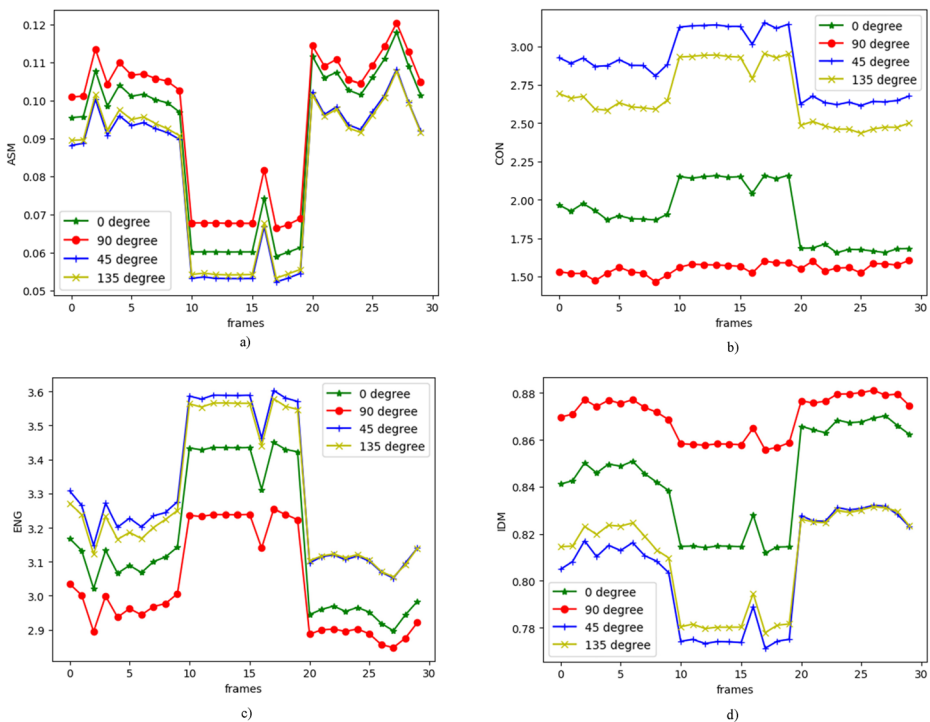


Fig. 4 Variation of eigenvalues of gray level co-occurrence matrix in different directions: **a** energy, **b** contrast, **c** entropy, **d** inverse difference moment

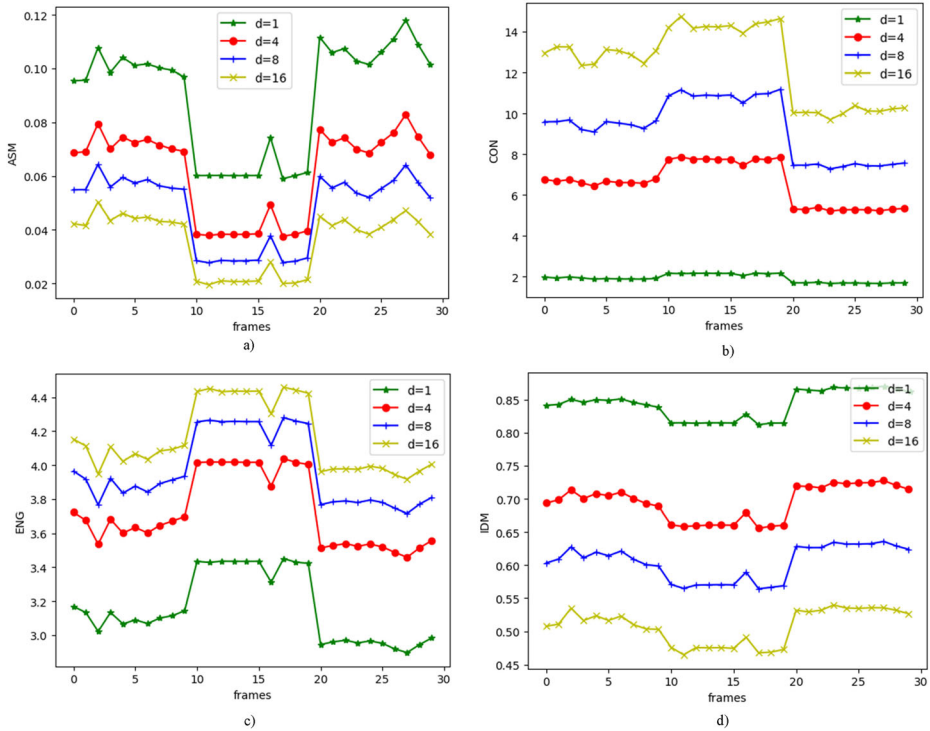


Fig. 5 The change of the feature quantity of the gray-level co-occurrence matrix when the distance is 1, 4, 8, 16: **a** energy, **b** contrast, **c** entropy, **d** inverse difference moment

complete. Therefore, on the basis of taking into account the amount of calculation and the integrity of texture information, this paper chooses d as 4 when calculating the gray level co-occurrence matrix.

3.3 Combining LBP and GLCM to extract image features in the bus

This paper uses the method of fusing LBP and GLCM to extract the texture features of the image in the bus. Specific steps are as follows:

- (1) Use the LBP Operator to calculate each pixel of the image in the bus to obtain the LBP image.
- (2) Calculate the gray level co-occurrence matrix of the LBP image obtained in the previous step, select the distance d as 4, and the direction angle θ as 0° , 45° and 90° , respectively, and calculate three gray level co-occurrence matrices.
- (3) Calculate the respective energy, contrast, entropy and inverse difference moment of three gray level co-occurrence matrices obtained in the second step and combine them to obtain a 12-dimensional texture feature vector for classification.

Take 100 sample images from each of the four types of crowding samples in the established bus image sample library as examples, follow the above steps to extract texture feature quantities and analyze the distribution of each feature quantity. Figures 6, 7, 8 and 9 are

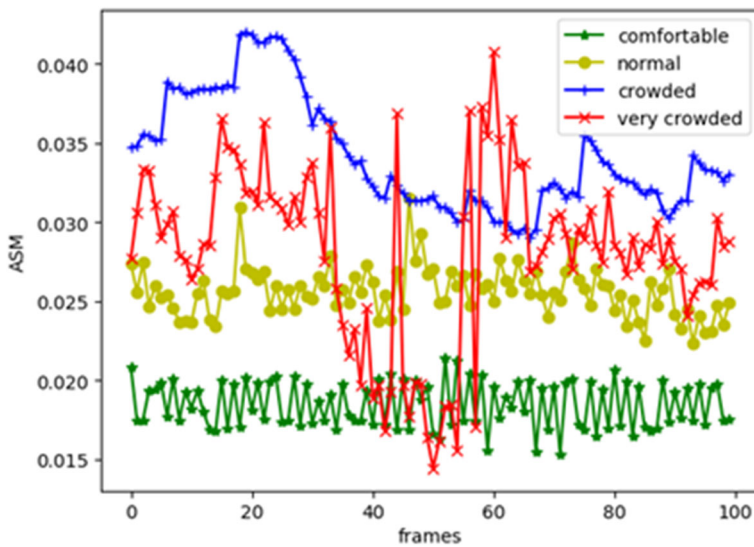


Fig. 6 Energy value curve

graphs showing the characteristic quantities of energy, contrast, entropy and inverse difference moment when $\theta = 0^\circ$.

In Fig. 6, analysis of the energy value curve shows that the energy value range of the sample image when the crowding is comfortable is [0.015, 0.021], and the average value is 0.018. When the crowding is normal, the sample image energy value range is [0.022, 0.032], and the average value is 0.026. When the crowding is crowded, the sample image energy value range is [0.029, 0.042], and the average value is 0.034. When the crowding is very crowded, the sample image energy value range is [0.014, 0.041], and the average value is 0.028. From

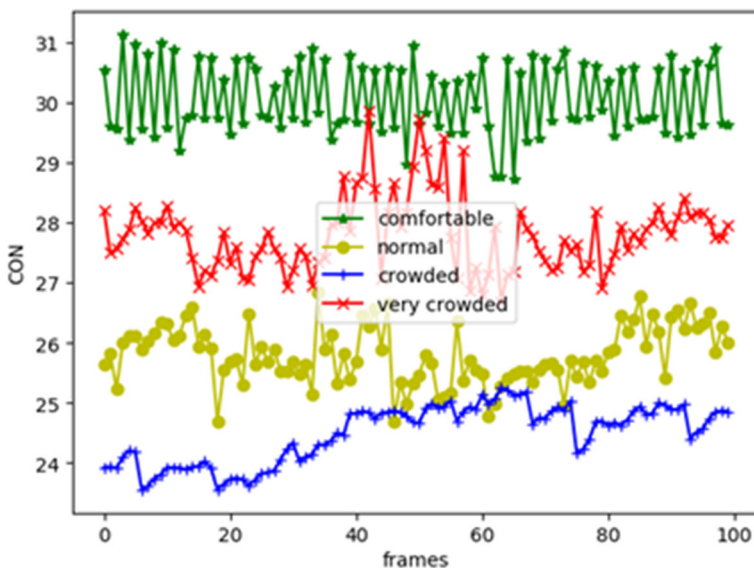


Fig. 7 Contrast curve

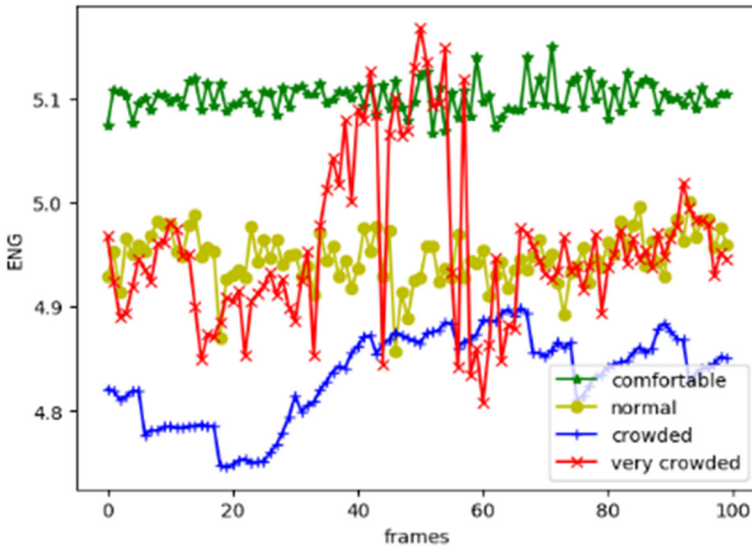


Fig. 8 Entropy curve

the above figure and the above analysis, it can be found that the energy value curve of the sample image when the crowding is comfortable, normal and crowded is obviously different, which can be easily distinguished. The energy value curve when the crowding is very crowded crosses the other three curves, but the overall trend of change is quite different, and the difference is more obvious after taking the average value.

Analyzing the contrast value curve shown in Fig. 7, it can be seen that when the crowding is comfortable, the contrast value range of the sample image is $[28.714, 31.128]$, and the average value is 30.055. When the crowding is normal, the contrast value range of the sample image is $[24.683, 26.831]$, and the average value is 25.799. When the crowding is crowded, the contrast

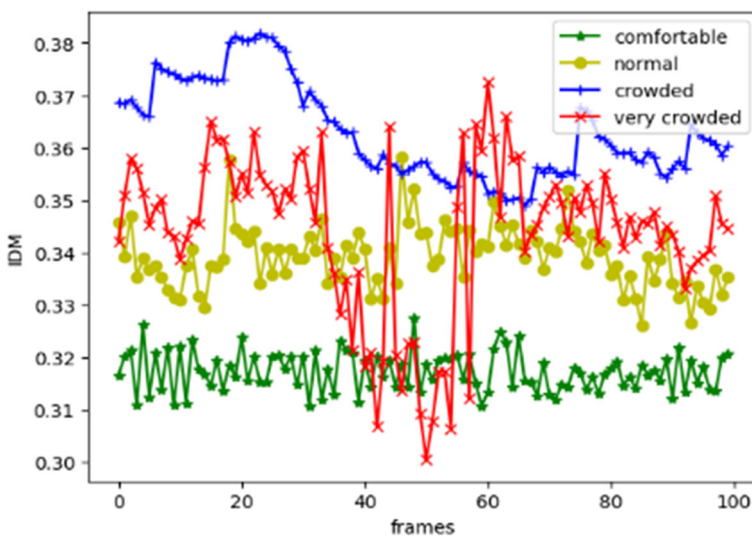


Fig. 9 Inverse difference moment curve

value range of the sample image is [23.555, 25.245], and the average value is 24.487. When the crowding is very crowded, the contrast value range of the sample image is [26.676, 29.864], and the average value is 27.796. From the above figure and the above analysis, it can be found that the contrast value curves of the four crowding figures are distinct and can be better distinguished.

In Fig. 8, the entropy curve shows that the entropy ranges of the sample image when the crowding is comfortable is [5.067, 5.149], and the average is 5.101. When the crowding is normal, the entropy range of the sample image is [4.858, 5.001], and the average value is 4.947. When the crowding is crowded, the entropy range of the sample image is [4.746, 4.898], and the average is 4.834. When the crowding is very crowded, the entropy range of the sample image is [4.808, 5.168], and the average is 4.958. From the above figure and analysis, it can be found that the entropy curves when the crowding is comfortable, normal and crowded are obviously different, which can be better distinguished. The entropy curve when the crowding is very crowded crosses the other three curves, but the overall change trend is quite different, and the difference is more obvious after taking the average value.

In Fig. 9, analyzing the graph of inverse difference moment shows that when the crowding is comfortable, the range of the inverse difference moment of the sample image is [0.311, 0.328], and the average value is 0.317. When the crowding is normal the range of the inverse difference moment of the sample image is [0.326, 0.358], and the average value is 0.339. When the crowding is crowded the range of the inverse difference moment of the sample image is [0.349, 0.382], and the average value is 0.363. When the crowding is very crowded the range of the inverse difference moment of the sample image is [0.300, 0.373], and the average value is 0.344. From the above figure and analysis, it can be found that the inverse difference moment curves when the crowding is comfortable, normal and crowded are obviously different, which can be better distinguished. The inverse difference moment curve when the crowding is very crowded crosses the other three curves, but the overall change trend is quite different, and the difference is more obvious after taking the average value.

4 Rough classification of crowding based on SVM

4.1 SVM-based rough classification algorithm design for crowding

The core idea of SVM is to map the input vector to a high-dimensional feature space through a certain nonlinear mapping (kernel function) and construct the optimal classification hyperplane to realize classification recognition [15]. While ensuring the accuracy of classification, the distance between both sides of the hyperplane can be maximized.

The actual classification problems are basically multi-classification problems. Assuming that an SVM classifier is designed between every two types of samples, $k(k-1)/2$ SVM classifiers need to be designed for k class samples [12]. Therefore, six support vector machine classifiers are needed for the rough classification of crowding. For an unknown sample, each classifier evaluates and counts its type, and the most statistical result is the type of test sample.

4.2 SVM parameter optimization

Since the penalty factor C and the radial basis kernel function parameter γ have a greater impact on the classification performance of SVM. This paper studies three SVM parameter

optimization algorithms such as grid search algorithm [13], particle swarm algorithm and genetic algorithm respectively, uses 500 training samples and 125 test samples of each type of crowding to test the performance of the three optimization algorithms, and selects the algorithm with the best optimization effect to build a rough classification model of crowding.

In the grid search algorithm, the value range of C and γ is selected as [2-10,210], the step length is taken as 0.5. Taking the accuracy of 5-fold cross-validation as the fitness function value, the highest fitness function value is 95.70%, and the corresponding optimal parameters are $C=90.51$ and $\gamma=0.0039$.

Particle swarm optimization algorithm [21, 27] is a swarm intelligence optimization algorithm that simulates the foraging behavior of birds. In the algorithm, the initial population number is set to 30, the inertia weight w is set to 1, the learning factors c_1 , c_2 are both set to 2, the maximum number of iterations is set to 100, and the variation range of C and γ is set to [0.001,100], take the accuracy of 5-fold cross-validation as the fitness function value of each particle, the highest fitness value is 95.40%, and the corresponding optimal parameter values are $C=94.75$, $\gamma=0.001$.

Genetic algorithm is a global adaptive probability search algorithm [2]. In the algorithm, the initial population number is set to 30; the crossover probability is 0.6; the mutation probability is set to 0.01; the maximum number of iterations is set to 100; the variation range of C and γ is set to [0,100], and the accuracy of 5-fold cross-validation is used as the fitness function value. The highest fitness value is 95.50%, and the corresponding optimal parameter value $C=83.98$, $\gamma=0.0064$.

Using three algorithms to optimize SVM parameters, the comparison of the test accuracy of the obtained crowding classification model is shown in Table 2. In the table, the SVM optimized by the three optimization algorithms are abbreviated as GSA-SVM, GA-SVM and PSO-SVM. It can be seen from the table that the classification accuracy rate is the highest when the genetic algorithm is used for optimization. Therefore, this paper selects the genetic algorithm to optimize the SVM parameters and uses the optimal combination of the obtained parameter (C , γ) to construct a rough classification model of crowding.

4.3 Construction of the rough classification model of crowding

The specific steps for establishing a rough classification model of crowding are as follows:

- (1) Choose five groups of samples as training sample with sizes of 200, 500, 1000, 1500, and 2000, and select the corresponding test samples according to the ratio of 8:2, which is a common choice [14], and extract the characteristics of the above samples.
- (2) Obtain the optimal parameters corresponding to the five groups of samples through genetic algorithm.
- (3) Use the optimal parameters and training samples obtained in the previous step to train to obtain five crowding classification models.

Table 2 Comparison of classification accuracy of three algorithms

	Best parameter combination	Test accuracy
GSA-SVM	(90.51, 0.0039)	93.00%
GA-SVM	(83.98, 0.0064)	93.20%
PSO-SVM	(94.75, 0.001)	91.60%

- (4) Test the five crowding classification models obtained in the previous step, and select the model with the highest test accuracy as the final crowding classification model.

Table 3 shows the optimal parameters and training accuracy of the five crowding classification models. It can be found that the training accuracy of the five classification models is high and the training accuracy is the highest when the training sample size of each type is 500.

In order to further test the performance of each classification model, five crowding classification models are tested with test samples. The results are shown in Table 4.

It can be found from Table 4 that when the training sample size is 500, the accuracy rate is the highest. Increasing and decreasing the training sample size will reduce the accuracy. When the training sample size is small, the SVM classifier cannot learn enough information, making its effect poor. When the training sample size is large, the computational complexity increases and the classifier learns too much useless information, which also affects the classification accuracy. Another important reason is that SVM is suitable for solving the problem of small sample classification, too much sample size makes it unable to show its unique advantages. From the above verification results and analysis, it can be seen that the performance of the SVM classifier is optimal when the training sample size is 500. Therefore, this paper uses the model trained when the training sample size is 500 as the final rough classification model of crowding.

4.4 Validation of crowding classification model

4.4.1 Analysis of rough classification test results of crowding in bus

This paper uses the video inside the bus to carry out the experimental verification of the rough classification of the crowding in bus. A frame is taken from every 25 frames of the video and sent to the SVM-based rough classification model of crowding, and the classification result is displayed in the video frame. The result is shown in Fig. 10.

The detection result of Fig. 10a is comfortable, indicating that the area per standing passenger get is greater than or equal to $1\text{m}^2/\text{person}$, passengers can move freely in the compartment, and most or all passengers have seats. The detection result of Fig. 10b is normal, indicating that at this time, the area of standing passengers is between $0.4\text{m}^2/\text{person}$ and $1\text{m}^2/\text{person}$. There is much space in the compartment, and there is basically no physical contact with standing passengers. The test result of Fig. 10c is crowded, indicating that the area of standing passengers is between 0.2 Between m^2/person and $0.4\text{m}^2/\text{person}$, standing passengers have more physical contact, and passengers are slightly crowded. The test result of Fig. 10d is crowded, indicating that the area of standing passengers is less than $0.2\text{m}^2/\text{person}$, and

Table 3 The training results of the rough classification model of crowding

Training sample size	Optimal parameter combination	Training accuracy
200	(59.65, 0.0031)	85.63%
500	(83.98, 0.0064)	95.50%
1000	(48.21, 0.001)	85.33%
1500	(98.82, 0.18)	85.65%
2000	(46.50, 0.15)	85.66%

Table 4 Performance test of five classification models

Training sample size	Test sample size	Comfortable accuracy	Normal accuracy	Crowded accuracy	Very crowded accuracy	Average accuracy
200	50	58%	64%	96%	92%	77.50%
500	125	72.80%	100%	100%	100%	93.20%
1000	250	85.60%	89.20%	79.60%	70%	81.10%
1500	375	89.60%	89.87%	86.13%	70.40%	84%
2000	500	87.60%	87.20%	74.40%	44%	73.30%

standing passengers frequently feel very crowded due to physical contact, which affects others getting on and off the bus.

Using 125 images of each type of test sample to test the optimal crowding classification model obtained in the previous section, the test results before parameter optimization are shown in Table 5, and the results after parameter optimization are shown in Table 6.

It can be seen from Table 5 that the average accuracy of the rough classification of crowding before parameter optimization is 78.80%, and the test accuracy of the four types of congestion are 60.00%, 72.00%, 97.60%, and 85.60% respectively. The classification effect is better when the crowding is crowded and very crowded, but the accuracy is lower when the crowding is comfortable and normal.

It can be seen from Table 6 that the accuracy of the SVM classifier after parameter optimization has improved. The average accuracy has increased from 78.80% to 93.20%, increased by 14.4%. In addition, the accuracy rate is increased to 100.00% when the crowding is normal, crowded, and very crowded, the accuracy rate when crowding is comfortable is improved less, from 60.00% to 72.80%, an increase of 12.80%. In general, the accuracy of

**Fig. 10** Rough classification results of crowding: **a** comfortable; **b** normal; **c** crowded; **d** very crowded

Table 5 Rough classification results of crowding before parameter optimization

	Comfortable	Normal	Crowded	Very crowded	Accuracy	Average accuracy
Comfortable	75	1	0	49	60%	78.80%
Normal	34	90	1	0	72%	
Crowded	3	0	122	0	97.60%	
Very crowded	0	18	0	107	85.60%	

crowding classification model after SVM parameter optimization has been greatly improved, and the rough classification has a better effect.

4.4.2 Analysis of results of fine classification test of crowding in bus

After completing the above-mentioned rough classification test of crowding, further research on the fine classification is done. First, selecting 500 training sample images for each type of crowding from the image sample library in the bus, a total of 3,000 images in six categories, and use these samples to train to obtain a GA-SVM-based crowding fine classification model. Then, select 125 images of each test sample for each type, and a total of 750 images for six types. These samples are used to test the fine classification model of crowding. The test results are shown in Table 7.

As can be seen from the above table, the accuracy of the fine classification of crowding based on GA-SVM is significantly lower than that of the rough classification of crowding, which is only 78.13%. Therefore, it is necessary to research on better methods to improve the effect of congestion fine classification.

5 Fine classification of crowding based on ResNet

5.1 Principle of ResNet

At present, deep convolutional neural networks have made a series of breakthroughs in the field of image classification. However, as the depth of the network continues to deepen, problems such as gradient disappearance and gradient explosion will occur, making it difficult for deep convolutional neural networks to converge. Although the standard initialization and the intermediate normalization layer have solved this problem well, as the network depth increases, a "degradation" problem occurs, that is, the accuracy rate drops rapidly after reaching saturation. In order to solve the "degradation" problem, He K et al. [11] proposed the ResNet (Residual Network) model, which does not directly fit the underlying mapping $H(x)$ but the residual mapping $F(x) = H(x) - x$. Therefore, the initial mapping is transformed

Table 6 Rough classification results of crowding after parameter optimization

	Comfortable	Normal	Crowded	Very crowded	Accuracy	Average accuracy
Comfortable	91	0	0	34	72.80%	93.20%
Normal	0	125	0	0	100%	
Crowded	0	0	125	0	100%	
Very crowded	7	0	0	125	100%	

Table 7 Fine classification results of crowding

	A	B	C	D	E	F	Accuracy	Average accuracy
A	70	0	0	0	6	39	56%	78.13%
B	0	119	1	4	1	0	95.20%	
C	0	0	62	13	50	0	49.60%	
D	1	0	5	116	3	0	92.80%	
E	2	0	0	0	105	18	84%	
F	12	0	0	0	0	113	90.40%	

into $F(x) + x$, which can be completed by shortcut connections in the network. The shortcut connection is a connection that can skip one or more layers. It is only used to implement identity mapping and add its output to the output of the overlay, as shown in Fig. 11.

The definition of the residual block is shown in Eq. (7).

$$y = F(x, \{W_i\}) + x \quad (7)$$

In the formula, x , y represent the input and output vectors respectively, and $F(x, \{W_i\})$ represents the residual mapping that needs to be learned.

Equation (7) requires that the dimensions of x and F must be the same. When the two dimensions are different, a linear projection needs to be added to match the dimensions, as shown in Eq. (8).

$$y = F(x, \{W_i\}) + W_s x \quad (8)$$

In the formula, W_s represents linear projection.

The residual block structure also contains a batch normalization layer [30], its main function is to solve the internal covariate offset problem in the training process of the convolutional neural network, that is, the input of each layer will change as the parameters of the previous layer change, which makes the training process more complicated and the training speed slow. In addition, batch normalization can also help solve the problem of gradient disappearance and gradient explosion, the main idea of which is to convert the mean and variance of each layer of input data into 0 and 1, respectively, and their positions are shown in Fig. 12.

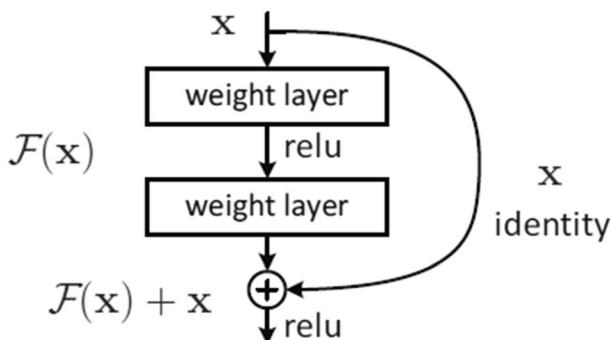
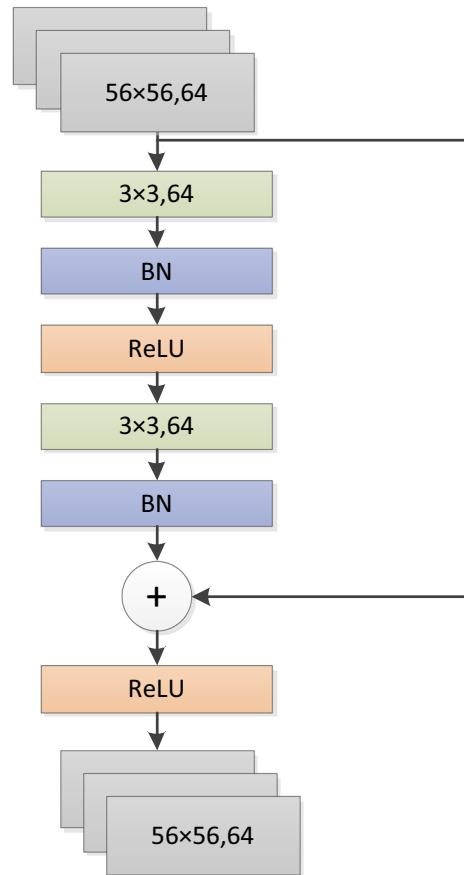
**Fig. 11** Ordinary residual block

Fig. 12 Model residual block structure diagram

5.2 Gradient descent optimization algorithm

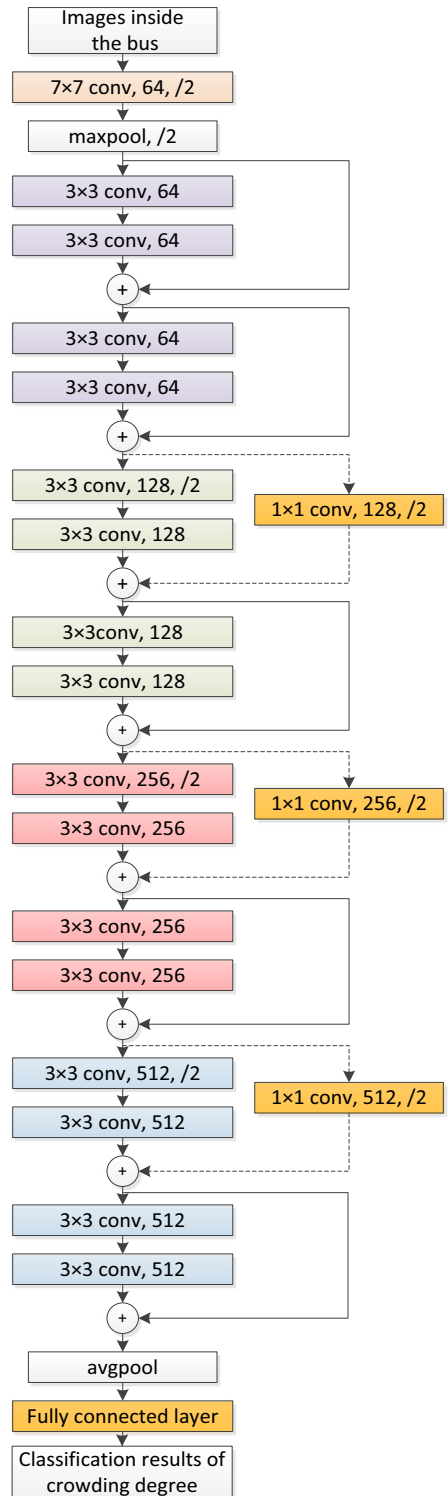
The commonly used gradient descent optimization algorithms are SGD, Momentum, Adagrad, RMSprop, Adadelata [26] and Adam [24]. In ResNet source paper, SGD optimization algorithm is used. RMSprop and Adadelata are the improved versions of Adagrad. Adam is the combination of momentum and RMSprop optimization algorithms. Therefore, this paper selects SGD, Adadelata and Adam optimization algorithms to compare.

The above three gradient descent optimization algorithms have their own characteristics and their performance in different application scenarios is also different. There is no final conclusion about which is better or worse. Therefore, in the follow-up of this paper, we will compare the effects of the three optimization algorithms through experiments and choose the most suitable optimization algorithm for the classification task of crowding in the bus.

5.3 Construction of a fine classification model of crowding

This paper builds a crowding fine classification model based on the ResNet-18 model and modifies the fully connected layer of the ResNet-18 model to adapt to the fine classification task. The detailed structure of the congestion fine classification model is shown in Fig. 13.

Fig. 13 Fine classification model of crowding



The input of the model in the above figure is the image inside the bus, the size is $224 \times 224 \times 3$, and the output is the fine classification of crowding result. The $n \times n$ indicates the size of the convolution kernel, numbers such as 64, 128, 256, 512, etc. indicate the number of convolution kernels, $/2$ indicates the step size, and it is not marked when the step size is 1. The solid line indicates that the number of input and output channels is the same, and Eq. (7) can be used directly for identity mapping; the dotted line indicates that the number of input and output channels is different, and the mapping of Eq. (8) is used to make the input and output dimensions the same, which is achieved by increasing the dimension by 1×1 convolution.

The size of the image sample is an important factor that affects the performance of the convolutional neural network. Since the number of samples in the established bus image sample library is small, this paper chooses three data enhancement methods: flip, color random jitter, and Gaussian blur to further expand the image sample size, select 3000 sample images of six types of congestion, and divide them according to the ratio of training set: verification set: test set=6:2:2, and uniformly scale the size of all sample images to $224 \times 224 \times 3$ to meet the model's requirements for the input image size. In addition, before inputting the sample image into the model, in order to reduce the computational cost, the image pixels need to be normalized, that is, the original pixel range of $[0,255]$ is normalized to the interval of $[0,1]$.

This paper uses Python language and based on the TensorFlow framework to build a fine classification model of crowding based on ResNet-18. The hyper parameter settings in the model training process are as follows: the batch size of the input data is set to 32, the number of loop iterations epoch of all data is set to 50. The learning rate in the SGD optimization algorithm is $\eta=0.01$; in the Adadelta optimization algorithm, $\rho=0.95$; $\varepsilon=1e-6$; in the Adam algorithm $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=1e-8$, $\eta=0.001$. The accuracy rate improves rapidly, which shows that the training effect of the congestion fine classification model based on ResNet is good.

5.4 Validation of the fine classification model of crowding

Using untrained in-bus video to verify the fine classification model of crowding in bus. Considering that the degree between adjacent video frames remains basically unchanged, in order to reduce the computational cost, a frame is taken every 25 frames, and it is sent to the ResNet-based fine classification model of crowding and the results are displayed in the video frame, which are shown in Fig. 14.

The detection result of Fig. 14a is A, indicating that the standing passenger area is greater than or equal to $1\text{m}^2/\text{person}$, the passengers can move freely, and most or all of the passengers have seats. The detection result of Fig. 14b is B, indicating that at this time, passenger's standing area is between $0.5\text{m}^2/\text{person}$ and $1\text{m}^2/\text{person}$, there is a distance between passengers and they are in a comfortable standing state. The test result of the Fig. 14c is C, indicating that the standing passenger area is between $0.4\text{m}^2/\text{person}$ and $0.5\text{m}^2/\text{person}$. Standing passengers have no physical contact, and they have basically the same personal space with sitting passengers. The test result of the Fig. 14d is D, indicating that the area of standing passengers is between $0.3\text{m}^2/\text{person}$ and $0.4\text{m}^2/\text{person}$, standing passengers occasionally have physical contact, and the personal space of them is smaller than that of sitting passengers. The test result of the Fig. 14e is E, indicating that the area of standing passengers is between $0.2\text{m}^2/\text{person}$ and $0.3\text{m}^2/\text{person}$, which is close to uncomfortable conditions, and standing passengers have frequent physical contact. The detection result of the Fig. 14f is F, which indicates that the standing passenger area is less than $0.2\text{m}^2/\text{person}$ at this time and is in an extremely crowded state.



Fig. 14 Fine classification results of crowding: **a** A; **b** B; **c** C; **d** D; **e** E; **f** F

Using 600 test samples of each type of congestion to test the accuracy of the fine classification model of crowding. The results of using three optimization algorithms show that the classification accuracy of Adam optimization algorithm is 96.22%, which is higher than the accuracy rate when the SGD and Adadelata optimization algorithms are used, and the accuracy rate for basic types of crowding is higher.

The fine classification results of crowding when using Adam optimization algorithm is shown in Table 8.

Table 8 Fine classification results of crowding when using Adam optimization algorithm

	A	B	C	D	E	F	Accuracy	Total accuracy
A	593	0	0	0	0	7	98.83%	96.22%
B	0	584	0	16	0	0	97.33%	
C	11	0	585	4	0	0	97.50%	
D	35	0	50	515	0	0	85.83%	
E	0	0	13	0	587	0	97.83%	
F	0	0	0	0	0	600	100%	

In general, the classification accuracy of the same model for samples with different crowding is different, and the classification accuracy of different models for samples with the same crowding are also different, the classification accuracy is always relatively high when the crowding is B, E, F, while the accuracy of other degrees is different. There are two main reasons for the above phenomenon: on the one hand, the difficulty of the samples at different congestion levels is different, which makes it easier for the model to learn the characteristics of some samples but difficult to learn the characteristics of other samples. Because the fine classification models of crowding trained by using different optimization algorithms have different learning capabilities for the same sample, the performance of the models obtained by different optimization algorithms is different when classifying the same samples.

Table 9 shows the comparison between the test accuracy of the fine classification model of crowding obtained by the Adam optimization algorithm and the GA-SVM method.

From the comparison results in the above table, it can be seen that the accuracy classification based on GA-SVM is only 78.13%, while the model based on ResNet reaches 96.22%, which is significantly higher than that based on GA-SVM method, it can better realize the fine classification of crowding in the bus.

6 Conclusion

This paper studied the crowding detection in bus based on the video images. Firstly, the image sample database is established based on the evaluation standard of crowding in bus. Then, LBP and GLCM are fused to extract the texture features of the image in bus. Then, a rough classification method of crowding based on SVM is proposed. In order to improve the accuracy of rough classification of crowding, the optimization effects of grid search algorithm, particle swarm optimization algorithm and genetic algorithm on SVM parameters are compared. Finally, in view of the unsatisfactory effect of SVM in the fine classification of crowding, this paper proposes a new method of fine classification of crowding in bus based on ResNet, and selects SGD, Adadelta and Adam algorithms to optimize the parameters of ResNet model, which effectively solves the problem of fine classification of crowding in bus.

Table 9 Comparison of GA-SVM and ResNet detection results

	A	B	C	D	E	F	Average accuracy
GA-SVM	56%	95.20%	49.60%	92.80%	84%	90.40%	78.13%
ResNet	98.83%	97.33%	97.50%	85.83%	97.83%	100%	96.22%

The verification results show that: (1) The crowding detection method in this paper can detect the majority of vehicles with experimental data, and the detection accuracy is high. (2) The texture feature extracted by the fusion of LBP and GLCM method can better characterize the crowding feature. (3) The performance of the rough classification model of crowding established after the optimization of SVM parameters is superior, and the parameter optimization effect of genetic algorithm is the best. The best parameters obtained by training can further improve the accuracy of rough classification of crowding of SVM. (4) The accuracy of the model based on ResNet is obviously better than that of SVM, which can better realize the task of the fine classification of crowding.

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